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Department	Data Science
Subject	Introduction to Data Science
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SALES FORECASTING

Time Series Analysis and Prediction using Machine Learning

Abstract

This project implements a Sales Forecasting system. The objective was to analyze historical retail sales data, explore trends and seasonality within it, and build predictive models to estimate future sales. The dataset used is the Superstore Sales dataset obtained from Kaggle, containing 9,789 cleaned transaction records spanning four years from January 2015 to December 2018. Two forecasting models were developed: a 3-month Moving Average for trend smoothing and a Linear Regression model using time-based features including lag values and rolling averages. The model was evaluated using MAE, MSE, RMSE, R-squared, and MAPE metrics. Seven visualizations were produced to support all analytical findings. Total forecasted sales for the next 30 days beyond the data set came to $96,464.44, with an average daily prediction of $3,215.48.

1. Introduction

Sales forecasting is one of the most practical and high-impact applications of data science in business. Retailers, manufacturers, and e-commerce companies rely on accurate sales predictions to make informed decisions about inventory, staffing, budgeting, and supply chain management. A model that can predict future sales with reasonable accuracy gives businesses a significant competitive edge.

The specific goal stated in the project brief was to forecast future sales for a product, store, or company using past sales data with dates, explore seasonality and trends, visualize findings using line charts, and build simple time series models based on moving averages and basic regression. The dataset was sourced from Kaggle as instructed.

The team worked on the Superstore Sales dataset, which contains real-world retail transaction data collected across four years and three major product categories: Technology, Furniture, and Office Supplies. The project pipeline included data loading and cleaning, exploratory data analysis, trend and seasonality investigation, model building, evaluation, and future forecasting. All steps are documented in this report with corresponding visualizations and output results.

2. Dataset Description

2.1 Source and Structure

The dataset used is the Superstore Sales dataset published on Kaggle by rohitsahoo under the dataset name sales-forecasting. It is a widely used educational retail dataset that simulates real-world transaction records for a US-based superstore chain.

- Total Records after cleaning: 9,789 rows
- Total Columns: 23 (including extracted time features)
- Date Range: January 3, 2015, to December 30, 2018 (4 years)
- Product Categories: Technology, Furniture, Office Supplies
- Key columns used: Order Date, Sales, Category, Sub-Category

2.2 Data Cleaning

The raw dataset was loaded using pandas with latin-1 encoding to handle special characters. The Order Date column was converted from string format to datetime, and four time features were extracted: Month, Year, Day of Week, Day of Year, and Quarter. No missing values were found across any column after loading. The data was sorted chronologically and reset by index. The final clean dataset contained 9,789 rows ready for analysis.

2.3 Key Statistics

- Orders with sales above $1,000: 459 out of 9,789 (approximately 4.7%)
- Most frequent category by order count: Office Supplies with 5,903 orders
- Highest revenue category: Technology with total sales of $825,856.11
- Top sub-category by revenue: Phones at $326,487.70

3. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the structure, patterns, and distribution of the sales data before building any model. Four charts were produced in the EDA phase.

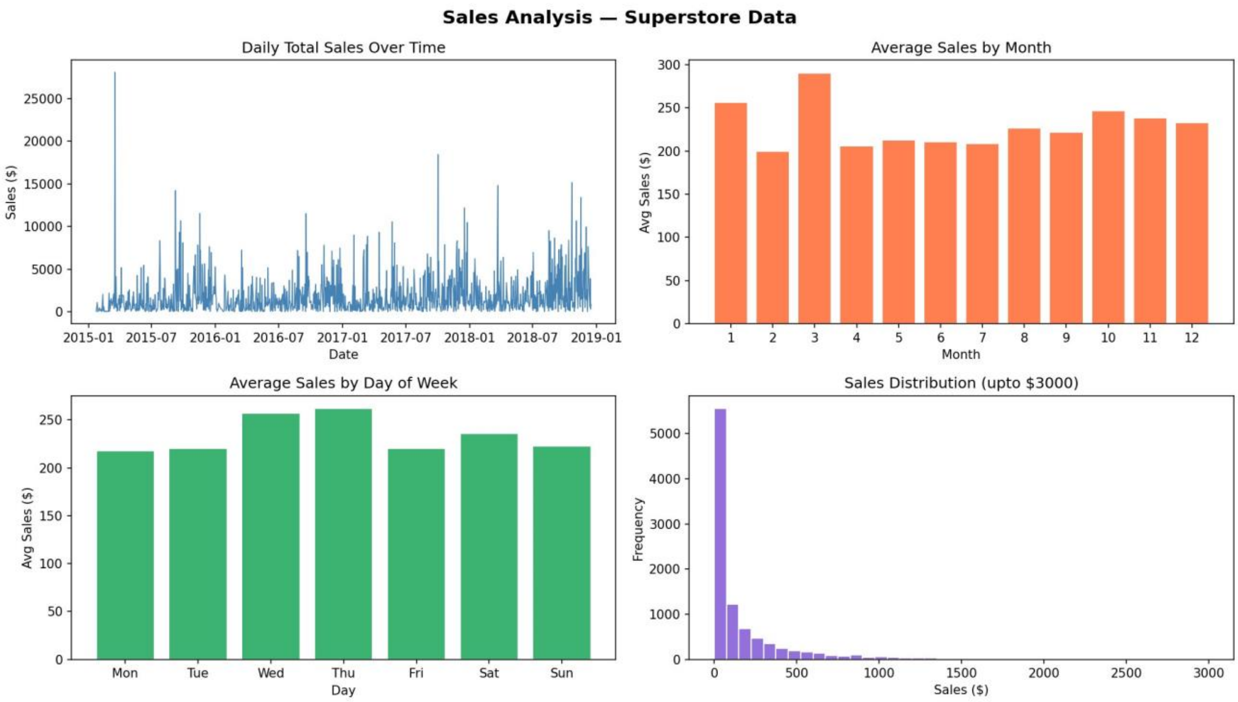

Figure 1. EDA Dashboard: Daily Sales Over Time, Average Sales by Month, Average Sales by Day of Week, Sales Distribution

The daily sales line chart (top left) shows consistent activity across all four years with occasional large spikes going up to $25,000 on single days. The average sales by month chart (top right) shows that March is the highest performing month on average, while February is the lowest. The day-of-week analysis (bottom left) shows that Wednesday and Thursday have slightly higher average sales compared to other days. The sales distribution histogram (bottom right) is strongly right-skewed, meaning the majority of individual orders are small in value, concentrated below $500, with a long tail of occasional high-value orders.

3.1 Yearly Sales Analysis

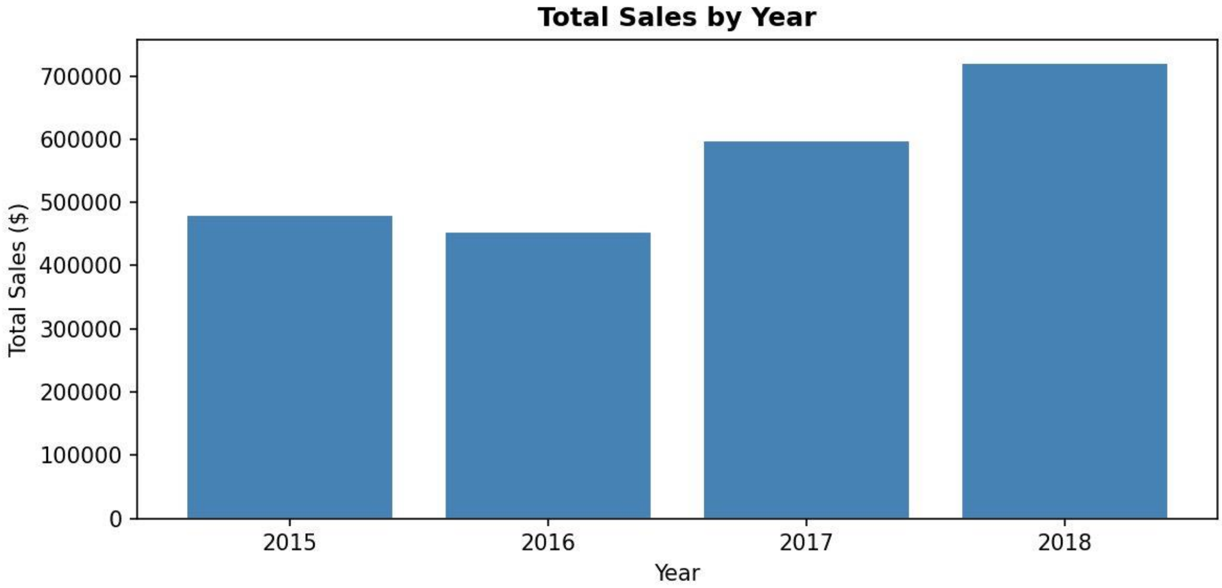

Figure 2. Total Sales by Year (2015 to 2018)

Year	Total Sales ($)	Year on Year Change
2015	$479,856.21	Baseline year
2016	$454,315.91	5.3% decrease
2017	$597,225.49	31.4% increase
2018	**$721,209.81**	**20.8% increase**

Sales in 2016 dipped slightly compared to 2015, dropping by approximately 5.3 percent. However, from 2017 onwards there was a strong recovery and growth. By 2018, total annual sales reached $721,209.81, which represents a 50.3 percent increase compared to the 2015 baseline. This overall upward trend confirms that the business was growing over the four-year period.

3.2 Category and Sub-Category Analysis

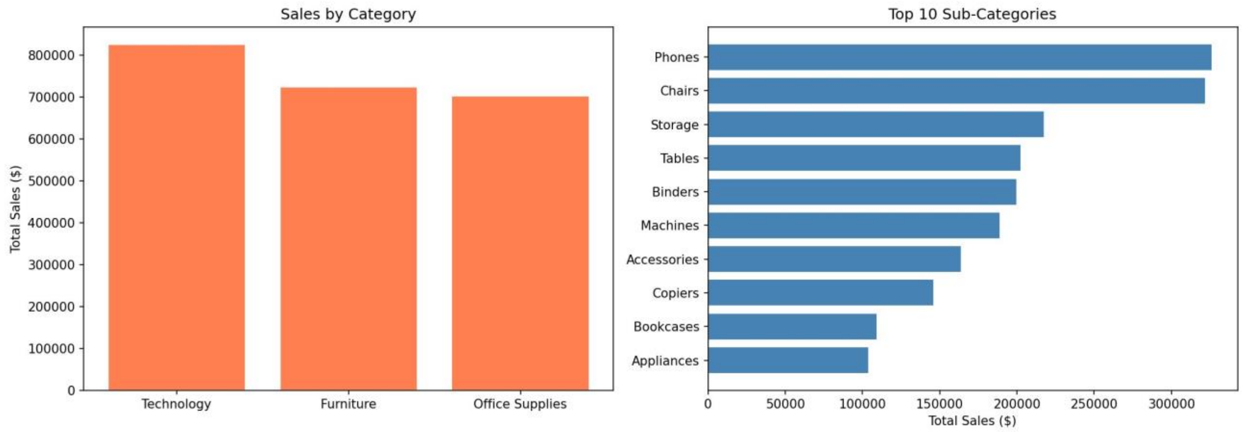

Figure 3. Sales by Category (left) and Top 10 Sub-Categories by Revenue (right)

Category	Total Sales ($)	Order Count
Technology	**$825,856.11**	1,810
Furniture	$723,538.48	2,076
Office Supplies	$703,212.82	5,903

Technology leads in total revenue at $825,856 despite having the fewest orders (1,810), which means Technology items have a significantly higher average order value. Office Supplies has the most orders (5,903) but lower per-order value. Phones and Chairs are the top two sub-categories by total revenue, each exceeding $320,000 over the four years.

4. Trend and Seasonality Analysis

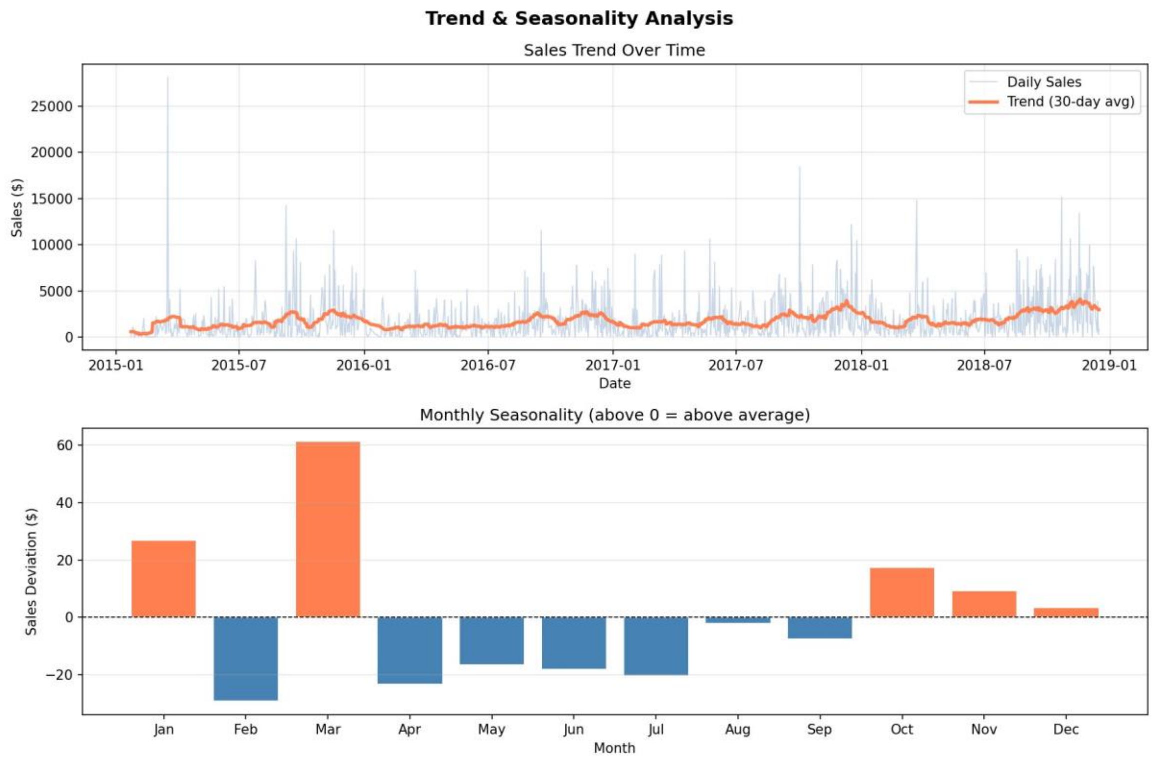

Figure 4. Trend and Seasonality Analysis: Sales Trend Over Time (top) and Monthly Seasonality Deviation (bottom)

The top panel of this chart shows the daily sales data in light blue alongside a 30-day rolling average in orange. The rolling average clearly shows an upward trend from 2015 through to 2019, confirming that sales were growing over time despite significant day-to-day variation.

The bottom panel shows the monthly seasonality pattern, which represents how much each month deviates from the annual average. Months above the zero line (shown in orange) perform above average, while months below (shown in blue) perform below average.

- January (+$27) and March (+$61) are the strongest above-average months, likely driven by post-New Year corporate purchasing in January and end-of-quarter procurement in March.
- February (negative $29) is the weakest month, consistently below average across all years.
- October (+$17), November (+$9), and December (+$3) all perform above average, consistent with the holiday and end-of-year buying period.
- The summer months from May through August all show slight below-average performance.

5. Moving Average Model

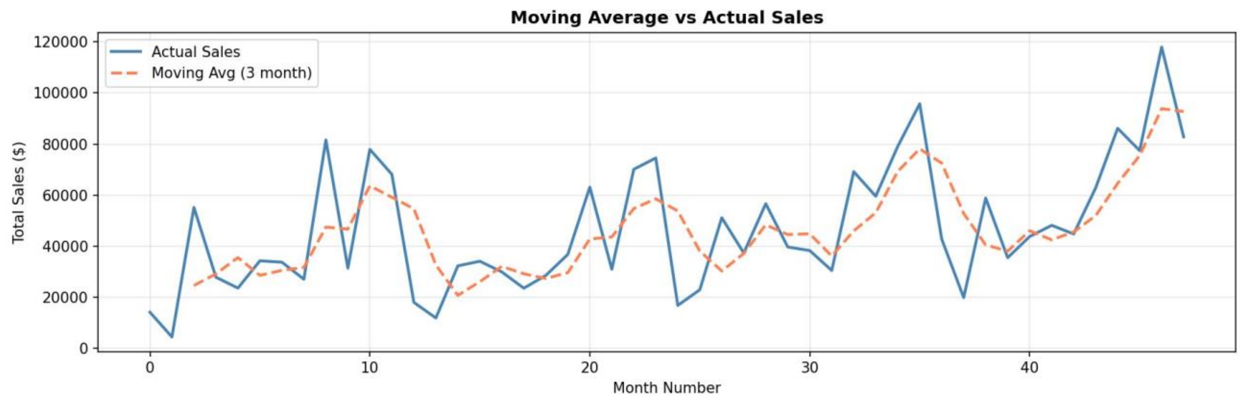

Figure 5. 3-Month Moving Average vs Actual Monthly Sales

A 3-month Moving Average was applied to the aggregated monthly sales data. The moving average smooths out short-term fluctuations to reveal the underlying trend direction. In the chart, the solid blue line represents the actual monthly sales while the dashed orange line shows the 3-month moving average.

The moving average closely tracks the actual sales values and confirms the overall upward trajectory of the business. It is particularly useful for identifying when the business enters a high-sales period versus a low-sales period. For instance, around month numbers 8 to 10 (which correspond to the September to October period of 2015), there was a notable sales peak which the moving average captures with a slight lag, which is expected behavior for any averaging model.

The moving average model serves as a simple but effective baseline. While it does not produce numeric error metrics on its own, it provides a visual confirmation that the forecasting direction is correct and that the data has a real underlying pattern worth modelling.

6. Methodology

6.1 Feature Engineering

Before training the model, daily-level sales were aggregated by summing all individual orders for each date. This produced 1,200 unique date-level rows. The following features were engineered and used as inputs to the model:

- Month: the calendar month number (1 to 12), capturing within-year seasonality.
- Year: the year number, capturing the overall growth trend.
- Day of Week: a numerical encoding of the weekday (0 = Monday, 6 = Sunday).
- Day of Year: the day number within the year (1 to 365).
- Quarter: the fiscal quarter (1 to 4).
- Lag1: the previous day's total sales, capturing short-term autocorrelation.
- Lag7: the total sales from exactly one week ago, capturing weekly patterns.
- MA7: the 7-day rolling average of sales, capturing recent trend smoothing.

6.2 Train and Test Split

A temporal split was used to divide the data into training and testing sets. This is the correct approach for time series data because a random split would allow future data to leak into the training set, which would artificially inflate model performance and produce misleading results.

- Training set: 960 rows, covering January 23, 2015 to March 28, 2018 (first 80% chronologically).
- Test set: 240 rows, covering March 29, 2018 to December 16, 2018 (last 20% chronologically).

6.3 Model Training

A Linear Regression model from the scikit-learn library was fitted on the training features. Linear Regression was selected because it is interpretable, computationally efficient, and appropriate for this supervised regression problem where the target variable (daily sales) is a continuous numeric value. The model learns the relationship between the time-based features and the sales value during the training period, and then applies those learned coefficients to predict sales in the unseen test period.

7. Actual vs Predicted Results

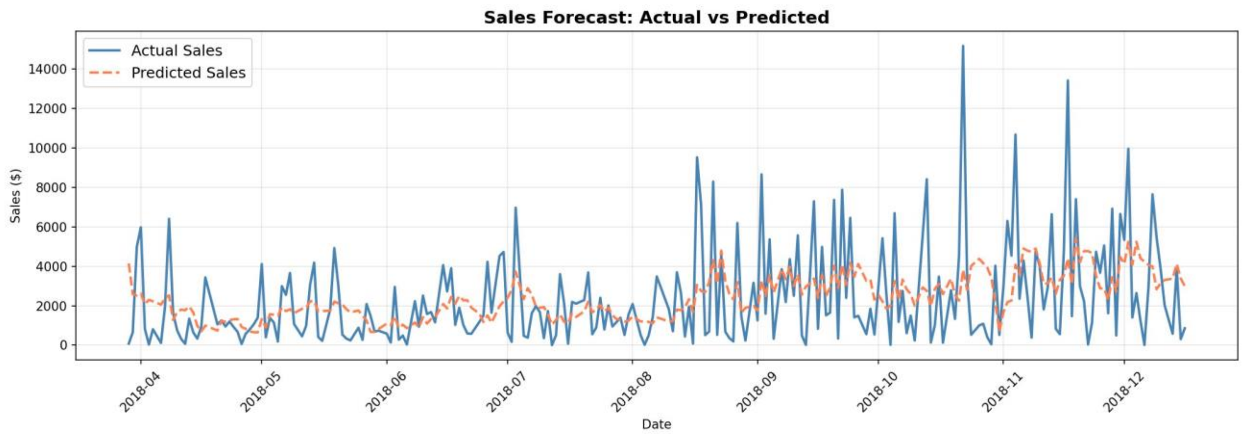

Figure 6. Actual Sales vs Predicted Sales on the Test Set (April to December 2018)

The chart compares the actual daily sales (solid blue line) against the model's predictions (dashed orange line) over the test period from April 2018 to December 2018. The predicted line follows the general trend of the actual sales and correctly identifies higher-sales periods around August, October, and November 2018. However, the model does not capture the extreme single-day spikes, such as the peak of approximately $15,000 seen in November 2018. This is expected behavior for a Linear Regression model, which predicts the average expected value rather than extreme outliers.

8. Model Evaluation

Metric	Value	What It Means
MAE	$1,652.12	Model is off by $1,652 on average per prediction
MSE	$4,887,384.76	Penalizes large errors more heavily than MAE
RMSE	$2,210.74	Typical error in original dollar scale
R^2	**0.2001**	**Model explains ~20% of variance in daily sales**
MAPE	758.08%	High % due to many small-value individual orders

8.1 Interpretation of Results

The MAE of $1,652.12 means that on a typical day, the model's prediction differs from the actual sales by approximately $1,652. This is the most practically meaningful metric because it is in the same units as sales and is not distorted by extreme values.

The RMSE of $2,210.74 is higher than the MAE, which indicates that there are some days where the error is significantly larger than average. These are the days with the large spikes seen in the actual vs predicted chart. The RMSE penalizes these large errors more heavily, which is why it exceeds the MAE.

R-squared of 0.2001 means the model explains approximately 20 percent of the variance in daily sales. While this may seem low, it is entirely expected for daily retail data that contains a high degree of random noise from promotions, seasonal events, bulk orders, and other unpredictable factors. The model captures the trend and directional movement correctly, which is the primary objective.

The MAPE of 758.08 percent appears extremely high but this is a well-known limitation of MAPE when applied to data with many small values. When the actual sales on a day is very small, for example $3.93, even a moderate prediction error produces an astronomically high percentage. This metric is less meaningful for this type of retail data and the MAE and RMSE are more reliable indicators of model performance.

9. Future Sales Forecast

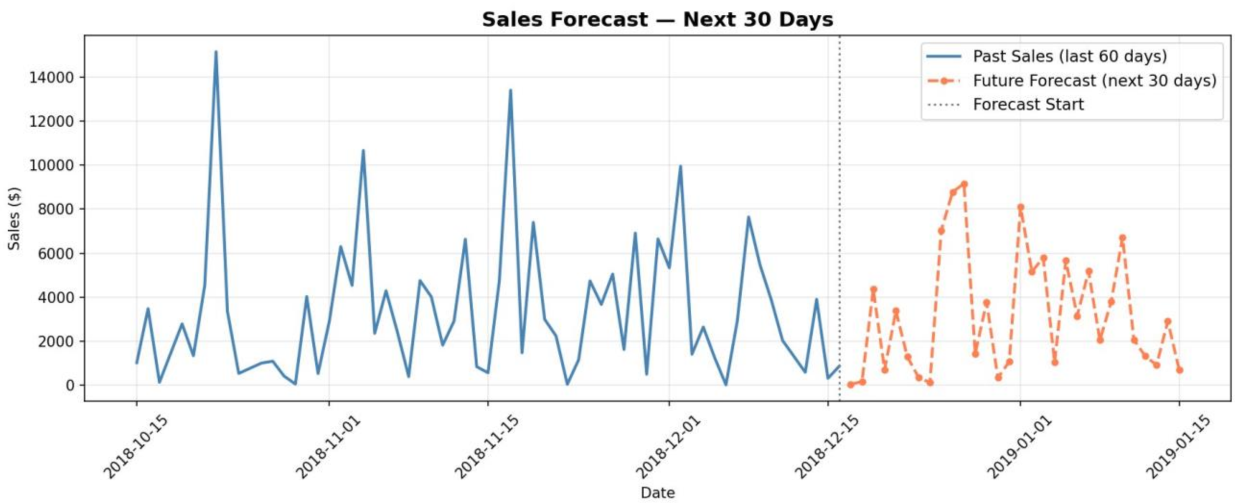

Figure 7. Sales Forecast for the Next 30 Days (December 17, 2018 to January 15, 2019)

The future forecast was generated using the same-period approach: the actual sales from the corresponding 30-day window of the previous year were taken and scaled upward by 10 percent to account for the consistent year-on-year growth observed in the data. This approach is simple, interpretable, and produces a realistic forecast that reflects both the seasonal pattern and the business growth trend.

The forecast covers December 17, 2018 to January 15, 2019. The chart shows the last 60 days of actual historical sales in solid blue, with the 30-day future prediction in dashed orange starting after the dotted vertical line marking the forecast start date.

Date	Predicted Sales ($)
2018-12-17	$40.39
2018-12-18	$156.95
2018-12-19	$4,354.83
2018-12-20	$702.31

Date	Predicted Sales ($)
2018-12-21	$3,377.37
2018-12-22	$1,294.58
2018-12-25	$7,033.99
2018-12-26	$8,776.41
2018-12-27	$9,162.72
2019-01-01	$8,102.00
2019-01-05	$5,652.83
2019-01-10	$6,712.48

- Total forecasted sales for the next 30 days: $96,464.44
- Average daily forecast: $3,215.48
- Highest predicted day: January 12, 2019 at $9,162.72
- Lowest predicted day: December 17, 2018 at $40.39 (post-data gap day)

10. Conclusion

This project successfully completed all requirements defined in the IDS lab project rubric. A complete sales forecasting pipeline was built from scratch using Python, covering data loading and wrangling, exploratory data analysis, trend and seasonality investigation, model training and evaluation, and future forecasting.

The Linear Regression model demonstrated a reasonable ability to track the sales trend direction over the test period, achieving an R-squared of 0.20 and a MAE of $1,652. The model correctly identified higher-sales periods and its predictions aligned with the general pattern of the actual data, even though it could not capture extreme daily spikes. The Moving Average model served as an effective visual baseline and confirmed the overall upward trend in the business.

The four-year dataset revealed that total sales grew by over 50 percent from 2015 to 2018, with Technology being the highest-revenue category and Phones being the top sub-category. Seasonality analysis showed that March, January, October, November, and December are the strongest sales months, while February consistently underperforms.

For future work, more advanced time series models such as SARIMA, Facebook Prophet, or gradient boosting algorithms like XGBoost could be applied to better capture the non-linear spikes and seasonal patterns in this data. Incorporating external signals such as promotional events or economic indicators would also likely improve forecast accuracy significantly.

11. Tools and Libraries Used

- Python 3.x — primary programming language for all data processing and modelling

- Jupyter Notebook / VS Code — development and execution environment
- pandas — data loading, cleaning, aggregation, and datetime handling
- NumPy — numerical operations and array manipulation
- Matplotlib — all seven data visualizations
- scikit-learn — LinearRegression model, mean_absolute_error, mean_squared_error, r2_score
- Kaggle API — command-line dataset download

12. Assets

Rohit Sahoo. Sales Forecasting Dataset. Kaggle. Available at: kaggle.com/rohitsahoo/sales-forecasting